

From the Dark Net to the Cloudy Data

Cloud Network Governance Guidelines

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Abstract—Cloud computing services are accessible from anywhere and used worldwide. There is plenty of information about concepts, architecture and models to deploy cloud systems. On the other hand, there is a lack of standardized information about utilization, contracts, and main performance metrics demanded by cloud services. Those data are usually private, non-structured and even customers commonly do not know exactly their contract clauses and demands. In this paper, we conducted a survey to investigate how the actual demands of customers are, in order to provide useful information for the specification of new models and to support decision making in cloud businesses. Based on the results obtained, we have identified the strong dependency of cloud services in the network level, and systematized an approach for cloud network governance to advance the state-of-the-practice in cloud governance and its relationship with business process management.

Keywords—cloud utilization survey; customer-centric monitoring; cloud network governance.

I. INTRODUCTION

Cloud computing introduced changes into companies' behavior regarding data processing and storage issues. Cloud computing providers offer outsourced computing services for their customers. As a result, new requirements emerged into business models, and the use of new strategies may be the competitive advantage to business [5].

Nevertheless, not only benefits come up with this new trend; other costs and complexity issues come along. New strategies for deploying IT Governance (i.e., planning, control, monitoring, and business) are required to promote transparency. Service providers must find a way to meet the demands of their customers, ensuring the principles of reliability, confidentiality, integrity, and availability [9].

There is a need for more specialized and regulatory monitoring tools, which can work within the provider's infrastructure, or outside it, letting customers know the current or past state of the services' performance. It is paramount to employ management accounting tools for both parties (i.e., customers and providers) [12]. Management accounting is the branch of accounting responsible for providing information to internal decision makers. It is also known as managerial accounting, or cost accounting [15].

In this paper, we conducted a survey on the characterization of cloud services' usage and, based on the results, we proposed a novel design to guide cloud governance focused on the network level, which we have called *cloud network governance*. We found out that circa of 188% of the Service Level Agreement (SLA) requirements are related to the network level (it is over 100% because a respondent may choose more than one option for this particular question). There is a need for an open Application Programming Interface (API) between cloud computing providers and customers for supporting management accounting on the customers' side as well. Only 15% of the providers under consideration make information available about SLAs' non-compliance metrics.

Despite SLA compliance metrics being a clear requirement, customers often do not have standardized access to them. Providers may benefit from it by over-allocating the unclaimed resources, or avoiding the payment of any owed penalties for not meeting SLAs. Beyond that, it is worth to disclose the SLA constraints contracted in practice. For instance, stakeholders will have information about the current performance related SLAs and, hence, they may have support to propose better economic models for cloud services. On the other side, providers may be aware of critical demands on investment in monitoring and management of assets to avoid financial losses.

There is a need to make network performance metrics available to customers, because such information is an essential business differential, as well as an important non-functional requirement usually missed by several cloud service providers. The knowledge of online performance metrics regarding services' execution contributes to foster selling and the deployment of cloud services.

The contributions of this work are a pilot study on the characterization of cloud services' usage, and the systematization of a cloud governance architecture (including control, monitoring, and management strategies), highlighting the importance of the network level to business process management. The appropriate sharing of network performance metrics can better support the trust relationship between customers and cloud service providers. The results contribute for guiding cloud service providers to establish priorities on

how to direct investments, and to support decision making by identifying both market (to attract new customers) and customers' demands (by increasing the trust degree on the service provider, and eventually leading to customer loyalty).

II. SURVEY ON THE CHARACTERIZATION OF CLOUD SERVICES

We have created an online survey based on the non-supervised research methodology for a pilot study. We performed a non-probabilistic sampling process based on *focus groups*, by initially selecting our own personal contacts and people belonging to regular e-mail lists. After that, we applied *snowball* sampling, by allowing participants to pass the survey onwards to their personal contacts [10]. Our investigation followed the steps below:

1. Identification of the cloud services used from the perspective of customers;
2. Identification of goals based on the GQM methodology, where the solution of each problem identified was described as objectives;
3. Identification of issues that covers all the objectives;
4. Identification of metrics for each question (a question might involve more than one metric);
5. Development of an online form with questions and sending to customers;
6. Analysis of the answers;
7. Validation of the questionnaire experiment;
8. Proposal of metrics for the design of *cloud network governance*;
9. Definition of a *baseline* for monitoring cloud service's quality indicators according to the obtained answers;
10. Assessment of results.

We developed a qualitative questionnaire following a 4-stage experimental process:

1. Definition of objectives;
2. Design of the experiment;
3. Execution of the experiment and data collection (descriptive statistics);
4. Analysis and interpretation of the collected data (inferential statistics).

A. Survey Objectives

The general objective of this survey is to characterize cloud computing service usage. More specifically, the specific objectives of the survey are the following:

1. To characterize the profile of cloud services' customers;
2. To characterize the utilization of cloud services;
3. To characterize technical requirements for cloud services as established by Service Level Agreements;
4. To identify which, if any, monitoring approach is being employed/offered by providers to their customers;
5. To characterize the levels of acceptance of cloud computing services by the customers.

We have categorized the survey questions into classes according to their specific goals. We summarized the

questions and goals within Table I by following the Goal, Question, Metrics (GQM) methodology [2]. We started with the definition of objectives that guided the creation of questions, and then the definition of at least one metric for each question.

TABLE I - CATEGORIZATION OF THE SURVEY QUESTIONS WITHIN THE OBJECTIVES.

Goals	Questions	Metrics
Objective 1: To characterize the profile of cloud services' customers	Question 1: What kind of company are you affiliated to (be as specific as possible)?	$M_{1,i}$ – % of customers affiliated to company type i
	Question 2: What is the area of expertise of your company?	$M_{2,i}$ – % of customers in the i -th area of expertise
Objective 2: To characterize the utilization of cloud services	Question 3: Choose the service(s) that best meet your needs.	$M_{3,i}$ – % of customers that need the service i
	Question 4: Do you need any cloud service that is not available for free?	$M_{4,i}$ – % of services not available for free
	Question 5: What kind of services do you use?	$M_{5,i}$ – % of customers that use the business model i
Objective 3: To characterize SLAs	Question 6: Do you have any contract to use cloud computing services?	$M_{6,i}$ – % if customers have a cloud service contract
	Question 7: If you pay for cloud services, such as Amazon EC2 or Windows Azure, do you know your cloud service contract clauses?	$M_{7,i}$ – % of customers that expect SLA constraint i
	Question 8: If so, which types of constraints were signed in your business contract?	$M_{8,i}$ – % of customers that have contracted the SLA constraint i
Objective 4: To characterize the current service monitoring approach (if any)	Question 9: Does your provider have any Web site at which you can take a look at your service performance?	$M_{9,i}$ – % of customers that use a performance monitor
	Question 10: If you answered yes to the previous question, does that Web site show failures and non-compliance with agreements regarding your service?	$M_{10,i}$ – % of customers that have access to non-compliance with SLA
Objective 5: To characterize the levels of acceptance of cloud computing services by the customers.	Question 11: How satisfied are you with your cloud computing service? Does it meet your technical requirements?	$M_{11,i}$ – % of customers that are satisfied as option i
	Question 12: Considering the time, effort and money you spent with your cloud service provider, how would you rate the overall value provided?	$M_{12,i}$ – % of customers that express cost-benefit as option i

B. Reliability and Validity

The assessment of the survey reliability used the *inter-observer reliability* approach. Initially we have validated the questions of the survey using *face validity*. We had two experts, who understand the topic and pretended they were filling the survey form to read the questionnaire, to criticize it, and to propose improvements [3]. We have employed the

content validity method to check out whether its content has all the information needed to achieve the goals. Each objective had its particular set of questions, covering the expected contents [10][14].

C. Descriptive Statistics

After cleaning the data inconsistencies, we obtained 67 answers, which are acceptable for a pilot study with 12 questions, since we met the goal of five respondents per question. We have coded the questionnaire, that is, we have mapped the responses into discrete numbers, and applied the appropriate strategy for analysis depending on the type of the question variable [3]. In descriptive statistics, we will describe the categorical and ordinal questions using pie or bar charts, and we will show measures of central tendency, such as the mean, median, and mode.

We have gathered answers from people affiliated to various companies/institutions (results in Fig. 1). The results for the company type show as the mode the *state-owned companies*. We have collected answers of companies from three economic sectors: education, services, and industry. Users from the education sector ranged over the categories of students, researchers, analysts, and professors, mostly employing cloud computing services to support their academic duties. The majority of the respondents come from the educational sector, 69%, followed by services (25%), and industry (6%), as depicted in Fig. 2.

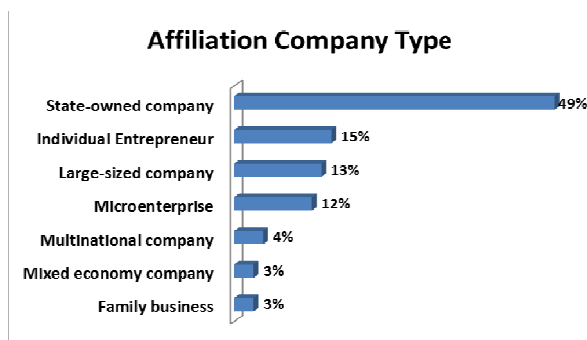


Fig. 1 - Pareto diagram of Question 1.

Google free-of-charge applications (e.g., Docs, YouTube, and Picasa) were the most popular among customers, used by 90% of them. Amazon Elastic Computing, which provides an infrastructure for high performance applications, is used by 36% of all users. Among the respondents, 46% declared they used other types of cloud services, such as Dropbox (45%), Apple iCloud (5%), and Evernote (5%). We present those results using Pareto diagrams (ordered frequency bar graphs) in Fig. 3. Note that the total may overpass 100%, because one respondent may choose more than one option. We have asked if customers needed to pay for cloud services. Most of them said no (61%), however a significant proportion of the customers declared they needed to paid cloud services (39%), as shown in Fig. 4.

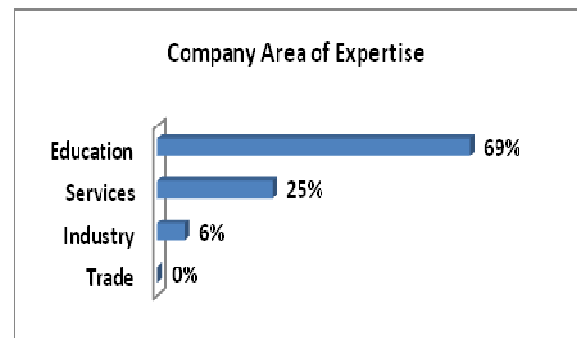


Fig. 2- Pareto diagram of Question 2.

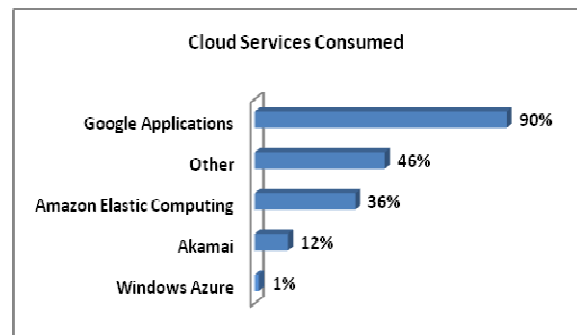


Fig. 3 - Pareto diagram of Question 3.



Fig. 4- Results obtained for Question 4.

Regarding the type of services, 78% of customers opted to use Software-as-a-Service (SaaS). In the second position came Infrastructure-as-a-Service (IaaS), being used by 69% of customers, while Platform-as-a-Service (PaaS) is used by 40% of them (one respondent may pick out more than one type), and 3% of customers did not know how to categorize cloud services, as presented in Fig. 5. Customers that contracted cloud services account for 34% of all customers (as shown in Fig. 6).

Only 18% of the total number of respondents has knowledge of the constraints stated in their cloud computing service contracts, as shown in Fig. 7. The SLAs negotiated between cloud service providers and customers may encompass many constraint types, or technical objectives.

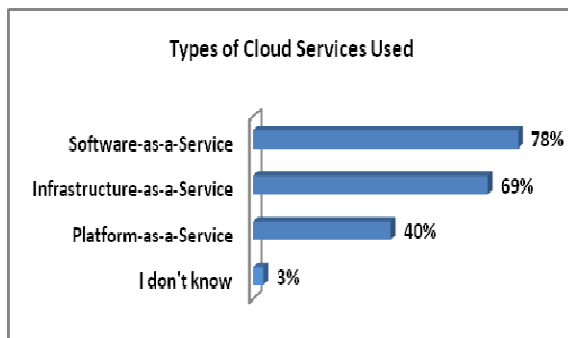


Fig. 5 - Pareto diagram of Question 5.

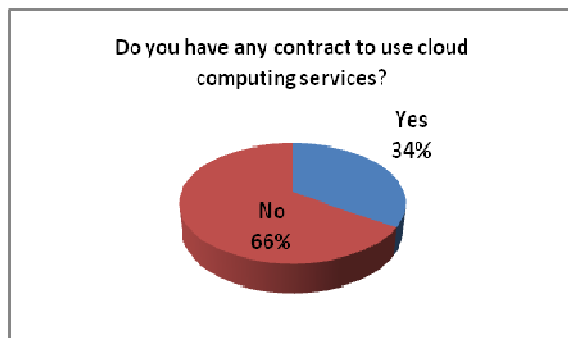


Fig. 6 - Results obtained for Question 6.

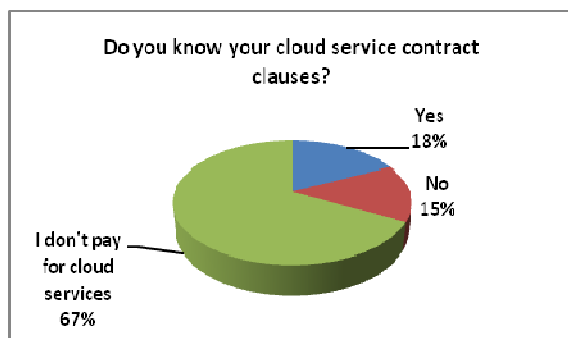


Fig. 7 - Results obtained for Question 7.

Some of the customers who declared to know their expected Service Level Objectives (SLOs) presented additionally the following performance requirements: bandwidth, delay, jitter, response time, availability, and time-to-repair from a fault. Those requirements were not a mandatory question, thus, not all the respondents specified their SLOs. Since one customer application may demand more than one SLO, the total may surpass 100%. When observing SLO types, the availability is the most frequent requirement with 18%, followed by network with 13%, and customer service with 6% (Fig. 8).

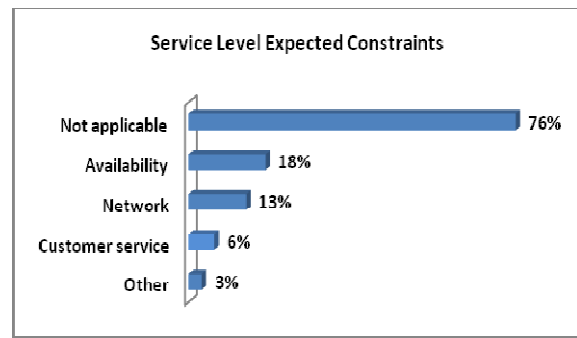


Fig. 8 - Pareto diagram of Question 8.

Albeit availability is as an application-level metric, it might also be classified as a network metric. Response time and security may also be monitored at network level. Usability issues generally involve the application level, such as the requirements regarding the graphical user interface; however, this requirement may also fit within the network level; e.g., DHCP (*Dynamic Host Configuration Protocol*) employed for automatically configuring machine names and addresses [13]. We have summarized the constraint types in Table II. The percentages correspond to the number of respondents that declared knowledge on SLA constraints. The total may overpass 100%, because one respondent may choose more than one option.

TABLE II - CLASSIFICATION OF THE CLAIMED SLA METRICS (QUESTION 8).

Metric	Type	Percentage
Availability	Network or Application	56%
Storage	Network or Application	38%
Response Time	Network or Application	25%
Delay	Network	19%
Bandwidth	Network	19%
Jitter	Network	13%
Security	Network or Application	13%
Usability	Network or Application	6%
Total		188%

A sum of 55% of the respondents do not dispose of any application to keep track of their services' performance metrics, leading to 45% that have access to a Web site containing such information, as depicted in Fig. 9. The survey pointed out that only 15% of the providers make information available about non-compliance with the service agreements (Fig. 10).

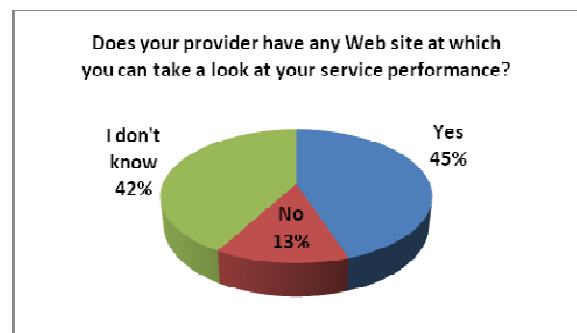


Fig. 9- Results obtained for Question 9.

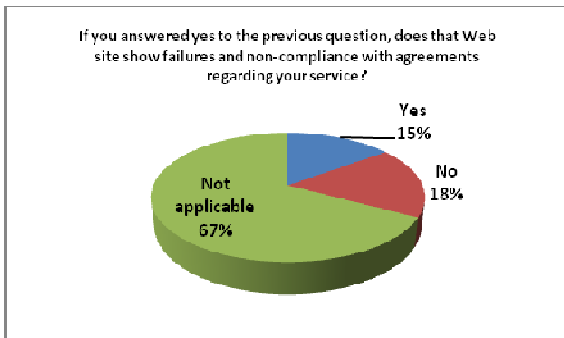


Fig. 10 - Results obtained for Question 10.

The customers on majority affirmed they are *very satisfied* with the meeting of technical requirements of the cloud services they use (Fig. 11). The mode corresponds to 60%.

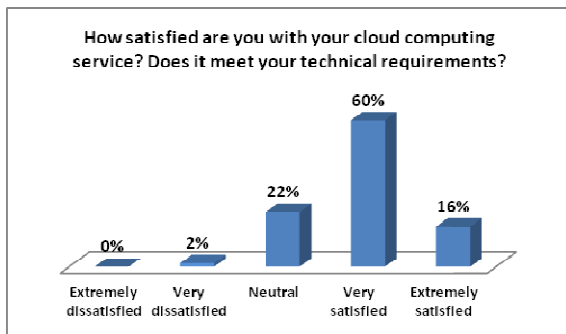


Fig. 11 - Likert Scale for Question 11.

When combining time, effort and money spent with cloud services, the customers' majority evaluated the services as *about what they expected* (57%), and 6% ranked the services as *worse than expected* (Fig. 12). This question had an additional box for customers to express any comment. Circa of 12% of the respondents made additional comments.

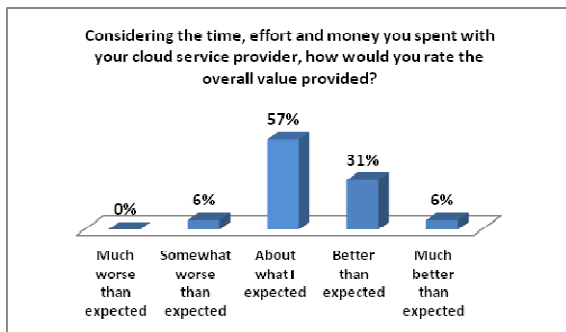


Fig. 12 - Likert Scale for Question 12.

The respondents additionally claimed for: not having bandwidth charges, even if bandwidth may be restricted after a certain amount; fine grained time slices; cloud database service to specific applications; private (intranet) cloud computing applications; services need to be as much transparent as possible, for the customer not to have to worry about infrastructure issues; the possibility to do advanced configurations in an easy way on specific domains and hosts; support for other manners of payment in some countries, e.g.

in Brazil; applications running in a browser impose limitations to the usage, then, cloud applications for desktop PCs need to have their own software, such as in mobile phones; there are compatibility problems between services, different data formats and protocols; it is difficult to perform local backup.

D. Inferential Statistics

We have performed a multivariate analysis in order to capture the correlations among all the answers for the questions, and to reduce dimensionality. We used Principal Component Analysis (PCA). In the PCA context, the questions are treated as *variables*, and the respondents are called *individuals* [7][11].

We conducted the PCA analysis to discover how each question correlates to the others, and to help us project those distances in a plot. The biplot projects the two main dimensions of components, and shows the correlation coefficients of each question to the dimensions. For instance, the biplot shows us that if the customer overall satisfaction is high, then the customer satisfaction with the technical requirements is also high, and that customers that contracted cloud services tend to not know their SLA clauses. We showed the biplot results in Fig. 13.

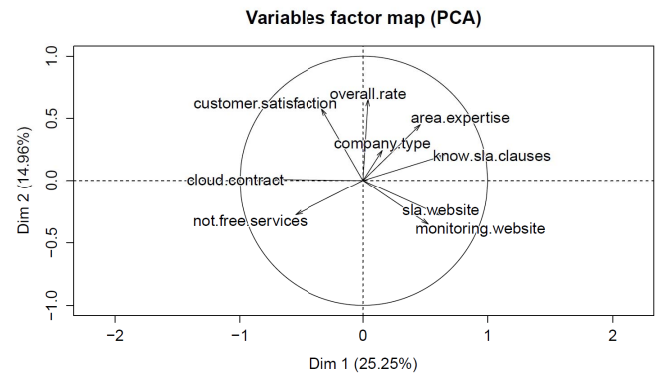


Fig. 13 – Biplot.

PCA contributes to reduce the dimensionality of the data by focusing on the components that had better explained the variability of the results. We shall retain the components that have eigenvalues greater than one. In our analysis, we found out three main components (see scree plot - Fig. 14). They explain together circa of 55% of the variability of the answers (Fig. 15). The component that better explains the variability of the first dimension is “cloud.contract” and corresponds to Question 7. Note that it has a negative correlation with the dimension (-0.7247). The question that has the highest (positive) correlation to the second dimension is the “overall.rate” (0.6538), representing Question 12. In addition, the “company.type” (Question 1) better correlates to the third dimension (0.8253), but not shown in the 2D plot. As a conclusion, Questions 7, 12 and 1 summarize the whole results, and may explain the majority of the answers.

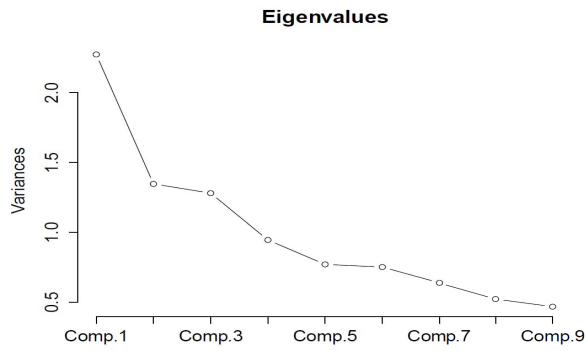


Fig. 14 – Scree plot.

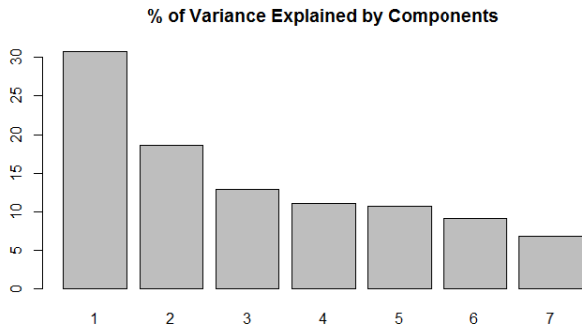


Fig. 15 – Percentage of variance explained by each component.

We have analyzed if there was statistical difference on the overall satisfaction from two categories of customers, the ones that have paid contracted cloud services, and that do not. Since we have a large proportion of the frequency of the answers summing less than five, then we will use the Fisher's exact test, because it provides a more accurate p -value. If the p -value for the statistical test is greater than 0.05, then we accept the null hypothesis that there is no significant difference between the question's categories. We got a p -value of 0.46, thus we accepted the alternative hypothesis. However, we noticed that it is a slight statistical difference.

We have plotted the overall satisfaction percentages for the customers that do not have a cloud contract (No), the ones that have (Yes), and for the total number of respondents (Total) in Fig. 16. Variable N means the number of samples, df is the degrees of freedom, and ϕ_c is the Cramér's Phi value (a.k.a. Cramér's V value) [4]. The ϕ_c statistic measures the intercorrelation of two discrete variables, and is applied to the goodness of fit of chi-squared models, $0 \leq \phi_c \leq 1$. If $\phi_c = 1$, then the two variables are equal. In our study, $\phi_c = 0.35$, which indicates that the intercorrelation of the satisfaction categories is moderate.

Finally, we checked the internal consistency of the questions using the test called Cronbach's α (CA). As a result, we obtained three factor loadings (components) from our survey questions that measures the correlation among questions. Questions measuring the same thing should load into the same factor.

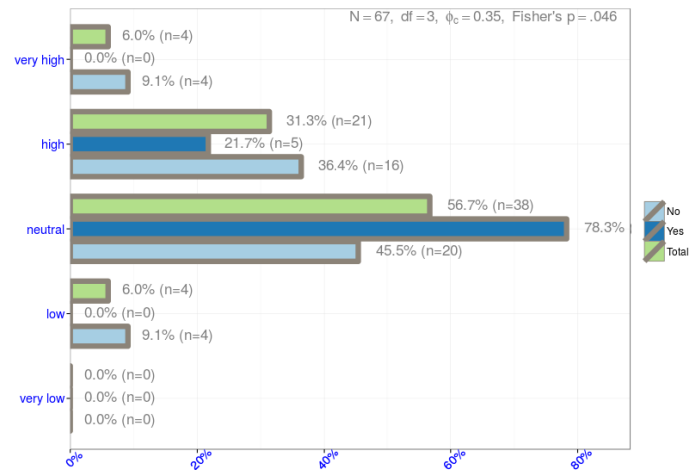


Fig. 16 – Overall satisfaction categorized by customers with and without cloud contract.

We found out a higher correlation of questions pertaining to the same objective, and then forming components with those questions, which validated how we mapped the questions into each objective. On the other side, the consistency test presented low values for CA, and consequent low internal consistency (*i.e.*, less than 0.6 in module) that indicates that the individuals have similarities. To improve internal consistency, we have deleted the highlighted questions from the experiment, and the components retain only the questions that have loading factors greater than 0.6. We may observe the CA analysis results in Table III.

TABLE III - INTERNAL CONSISTENCY.

Principal Component Analysis (with varimax rotation)			
Question	Component 1	Component 2	Component 3
area.expertise	-0.29	0.07	-0.77
cloud.contract	0.74	0.18	0.00
company.type	0.14	-0.11	-0.86
customer.satisfaction	0.06	0.72	0.12
know.sla.clauses	-0.72	0.03	-0.07
monitoring.website	-0.42	-0.47	0.05
not.free.services	0.65	-0.14	0.06
overall.rate	-0.26	0.63	-0.11
sla.website	-0.38	-0.48	-0.12
Cronbach's α	-1.00	-0.05	0.38

III. PROPOSAL FOR CLOUD NETWORK GOVERNANCE

The Brazilian Corporate Governance Institute (IBGC) define governance as follows: “*Corporate Governance is the system whereby organizations are run, overseen and incentivized. It involves relationships between the shareholders, the Board of Directors, the Officers and oversight bodies. Good corporate governance practices convert principles into objective recommendations, aligning interests with the purpose of preserving and enhancing the organization's value, facilitating its access to capital and contributing to its longevity.*” [8].

Cloud computing systems are strongly dependent on the network. Such network dependency is evident in the survey results, where 188% of the SLA constraints have relationship to the network level. Therefore, we advocate that it is of major importance to give special attention to implement a methodology to monitor the network, manage SLA requirements, and promote information available for the management accounting for both customers and providers.

IT governance covers four main phases: (i) identification of needs; (ii) solution envisioning; (iii) solution planning; and (iv) solution implementation [1][6]. When focusing on the cloud network governance, in the first phase we gather the business strategies. After that, we define network traffic capture and preprocessing for filtering out the relevant packets, or flows, related to cloud services. The second phase consists on monitoring SLOs as expected from SLAs for each contracted service. After that, we perform the SLA traffic compliance, triggering non-compliance events according to previously defined policies. Phase 3 corresponds to the processing of anomalies and other control events, as well as the planning of actions. The management accounting actions by both customers and providers occurs in the fourth phase. Those steps contribute to foster the credibility of the charging process. We summarized the basic principles of the cloud governance with focus on the network level in Fig. 17.

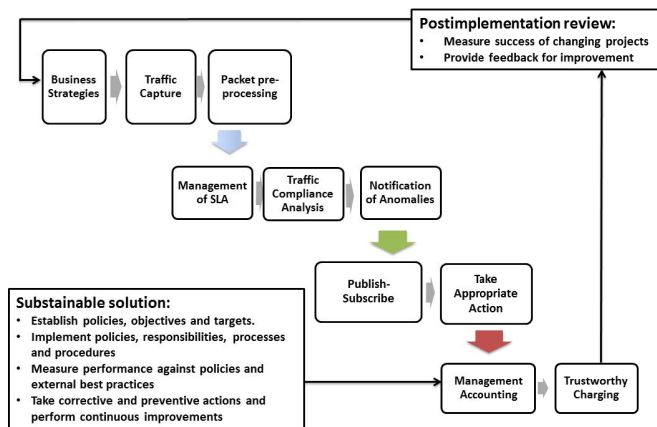


Fig. 17 - Cloud network governance under COBIT perspective [6].

The cloud network governance is detailed as follows:

- **Business strategies:** the business needs will be raised with focus on network traffic. Only the traffic packets that are relevant to the business are going to be captured, monitored and analyzed;
- **Traffic capture:** the traffic is captured through port mirroring;
- **Packet pre-processing:**
 - **Traffic filtering:** monitoring only the packets that indeed require to be monitored;
 - **Metrics computation:** the network metrics stated at the SLA are computed for the monitored instances;

- **Service level agreement management:** this module is responsible for storing the SLA constraints for each service. SLAs ought to be received via a *publish-subscribe* (P-S) *control channel*;
- **Traffic compliance analysis:** this module is responsible for verifying if the measured metrics actually meet the objectives (SLOs); in case an SLA violation is found, a publisher-subscriber notification channel is employed for sending an alert message. All customers that want to have access to performance metrics may subscribe to the notification channel;
- **Notification of anomalies:** the notification interface communicates with the *P-S notification channel* by receiving messages regarding services performance in real time;
- **Management accounting:** besides being a front-end for viewing traffic metrics and SLA violations in real time, it may serve as a tool for management accounting of cloud computing services. Beyond that, since the P-S notification channel is a flexible solution, that information may also be used by other service provider components, in order to allocate virtual machines (VM) resources into physical hosts, estimate costs, and even to support decision-making based on its own management accounting expectations;
- **Trustworthy charging:** the charging process changed from a one-sided process to a process where the client have information to disprove the final charges, if needed.

We have systemized the cloud network governance into the life cycle of business process management [16] that comprehends the four main phases previously listed. We ask special attention to the notifications of anomalies for both customer and provider parties. It is a business differential and contributes to improve trust relationship, especially regarding charging issues. The creation of a fully functional business process involves five steps, often referred as the BPM life cycle: design, modeling, execution, monitoring and optimization. Each of these five steps represents an important development phase in implementing a successful process solution.

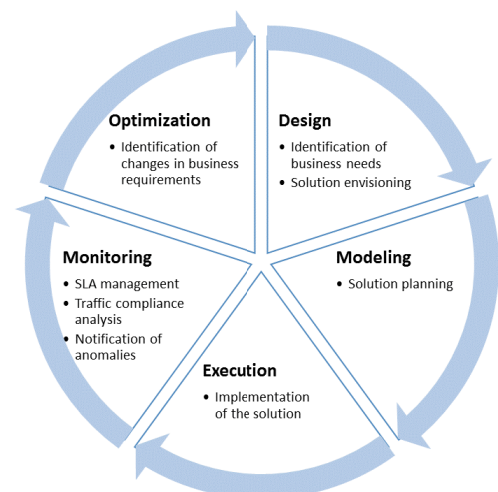


Fig. 18 – Life cycle of Business Process Management aligned with Cloud Network Governance, adapted from [16].

- **Design:** the first step consists of activities that define the business process: identifying high-level activities, discussing possible organizational changes, defining service level agreements, and specifying process details such as actors, notifications, and escalations. In such step, we identify *business needs* and *solution envisioning*.
- **Modeling:** in this step, the business process is fully specified and validated. The process flow is formalized, for example, by using BPMN. Additionally, process variables are defined and *candidate cloud services* that will be used to execute an activity are *identified and contracted*.
- **Execution:** the *business process model* is implemented in a business process application, often using a business process management system (BPMS). We need to add technical details to the business process before *execute* it.
- **Monitoring:** the processes are monitored for business goals that are assured by metrics defined in the SLA. Examples of metrics are availability, delay and response time. In this step, the cloud service provider performs the traffic compliance and if it *detects anomalies*, it should *notify the consumer*, which, in its turns, *verifies the level of acceptance* of the service quality.
- **Optimization:** based on new insights, changing business requirements, and monitoring results, the implemented business processes will need to be optimized. When the optimization phase is done, the business process goes into the design phase again and the cycle is completed.

The BPM life cycle shows that implementing business processes is an ongoing process due to the ever-changing business environment and need for optimization. How long it takes to walk through all five steps of the BPM life cycle depends greatly on the business environment and the ability of a business to execute. In cloud computing network environment, it is a challenging task to perform real time traffic analysis.

In order to promote the use of cloud network governance, we have developed in [12] a network monitoring architecture called TADE, which is an open source tool available in <http://jitclouds.lsd.ufcg.edu.br>.

IV. CONCLUSIONS

Cloud governance strategies focused on the network level helps to aggregate value to the business with acceptable risk and responsibility, contributing to encourage customers to buy cloud services and consequently increase the customers' trust on providers. The results reinforce the importance of transparency in the network service and charging process. Only 15% of the providers mentioned in the survey make information available about non-compliance with the service agreements, and the great majority of all customers have no knowledge about SLAs' constraints.

Cloud computing SLA monitoring is heavily network dependent. On the other hand, network traffic monitoring at wire speed in gigabit backbones is very complex. There is a demand for a complex orchestration of events when dealing with aggregated traffic, in order to filter out application flows, and verify SLA's violation.

Customers need to trust on cloud dependability. It is interesting to providers to increase the customers' trust degree in cloud computing services. We argue that making monitored data available to the customers and providing tools for SLO estimation, besides promoting guidance to negotiation of SLAs, will help to consolidate cloud services into critical information-dependent economic sectors.

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